How much dissipation can a multiscale layer cascade carry?

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2 November 2026

Preprint · v0.09, 2 November 2026 · Licence CC BY 4.0
CAOS Research open-problems programme · problem `navier-stokes`
Version DOI 10.5281/zenodo.23890775
Concept DOI (always latest) 10.5281/zenodo.22820520
Not peer reviewed. Source code, data and computational records:
github.com/fsantibanezleal/CAOS_RESEARCH

Abstract

Multiscale layer constructions produce finite-time blowup for several fluid equations: for the incompressible porous media equation, for the two-dimensional Boussinesq system, for three-dimensional Euler, and, with fractional dissipation $|\nabla|^\alpha$ of order $\alpha < (22 - 8\sqrt{7})/9$, for the forced Navier-Stokes equations. The exponent in the last of these is small, and it is natural to ask what limits it.

We give two upper bounds on the dissipation exponent such constructions can carry. The first is independent of every design constant: a cascade whose layers grow at rate $\sqrt{A}$ on a background gradient A cannot exceed $\alpha = 1/2$, and one whose layers grow at rate A cannot exceed $\alpha = 1$, both in the $|\nabla|^\alpha$ convention in which classical viscosity is $\alpha = 2$. Neither mechanism reaches classical viscosity, by factors of four and two respectively. The second bound applies to the smooth-forcing designs used for the inviscid Boussinesq and Euler results: both are of the pendulum class, and transcribing the exponent content of their force estimates we show that they certify dissipation only up to $\alpha \le 5.3 \times 10^{-4}$ (Boussinesq) and $\alpha \le 6 \times 10^{-8}$ (Euler) at the published choices, and only up to $\alpha \le 9.0 \times 10^{-3}$ and $\alpha \le 7.5 \times 10^{-3}$ when every constant of the design is retuned in its favour, at least ten times below the exponent already achieved with a force of finite regularity. In every reading the ceiling is highest when the fewest derivatives of the force are controlled.

Both bounds follow from the amplitude system of the construction together with one additional ingredient, the effect of fractional dissipation on a single layer, which we derive and then verify against direct simulation of the nonlinear equations, including the effect of the spatial localization the constructions use. We also reconstruct the exponent budget behind the published hypodissipative threshold, identify the constraint that determines it, and show that the admissible frequency ratios form an interval that closes exactly at that threshold, so that cascades with geometrically growing frequencies are excluded at every positive dissipation.

1 The question

Let θ and u solve the forced two-dimensional Boussinesq system, or let u solve the forced three-dimensional Navier-Stokes equations with fractional dissipation. Multiscale constructions in the

style introduced by Córdoba and Martínez-Zoroa build blowup by stacking oscillatory layers: each layer grows on the gradient deposited by its predecessors and hands a larger gradient to its successor, with infinitely many stages compressed into finite time.

Dissipation opposes this directly. A layer of frequency λ is damped at rate $\nu\lambda^{2\alpha}$ in the convention $-\nu(-\Delta)^\alpha$, while its growth rate is set by the background gradient alone. Frequency therefore buys gradient but not growth, and the question of how much dissipation a cascade tolerates becomes a question about the exponents relating amplitude, frequency and gradient. This paper answers it at two levels: for the mechanism classes (Section 3) and for two published designs (Section 4).

Conventions. We write dissipation as $-\nu(-\Delta)^\alpha$, so classical viscosity is $\alpha = 1$; results quoted from the literature in the $|\nabla|^\alpha$ convention have twice our exponent, and we state which convention is in use whenever a number appears. The hypodissipative threshold $(22-8\sqrt{7})/9 = 0.09267$ of [3], stated in the $|\nabla|^\alpha$ convention, is 0.04633 in ours.

2 The layer mechanism and its dissipative extension

On a background that is affine near the origin, $u_{\text{old}} = D(t)x$ and $\theta_{\text{old}} = G(t)\cdot x$, a temperature wave and a vorticity wave sharing one phase $s = \lambda\zeta(t)\cdot x$, with $\vartheta = \Theta F(s)$, $\varpi = \Omega F'(s)$ and $\psi = \Omega P(s)/(\lambda^2|\zeta|^2)$, satisfy a closed system of amplitude equations,

$$\dot\zeta = -D^T\zeta, \qquad \dot\Theta = -\frac{J\zeta\cdot G}{\lambda|\zeta|^2}\,\Omega, \qquad \dot\Omega = \lambda\,\zeta_1\,\Theta, \tag{1}$$

where $J(x_1,x_2) = (-x_2,x_1)$. The wave velocity is perpendicular to ζ while both wave gradients are parallel to it, so the wave does not advect itself and the ansatz is exact where the envelope is one and the profile is linear. In the frozen configuration $D = 0$, $G = -Ae_2$, $\zeta = re(\varphi)$, the growth rate is $\sqrt{A}\sin\varphi$: it carries no λ, because the frequency in $\dot\Omega$ cancels the reciprocal frequency in $\dot\Theta$. The gradient the layer deposits is governed by the amplification identity

$$\nabla\vartheta(0,t) = \lambda\Theta\zeta, \tag{2}$$

which is why a short wave can carry a small amplitude and a large gradient, and is what makes the cascade possible.

Dissipation. Adding $-\nu(-\Delta)^\alpha$ to both equations leaves the two geometric cancellations untouched and is diagonal on the phase, since $(-\Delta)^\alpha F(\lambda\zeta\cdot x) = (\lambda|\zeta|)^{2\alpha}F(\lambda\zeta\cdot x)$. The amplitude system therefore acquires one damping term per amplitude,

$$\dot\Theta = -\frac{J\zeta\cdot G}{\lambda|\zeta|^2}\,\Omega - \nu(\lambda|\zeta|)^{2\alpha}\Theta, \qquad \dot\Omega = \lambda\zeta_1\Theta - \nu(\lambda|\zeta|)^{2\alpha}\Omega, \tag{3}$$

with eigenvalues $-\nu(\lambda r)^{2\alpha} \pm \sqrt{A}\sin\varphi$ in the frozen configuration. Growth at frequency λ on background gradient A therefore requires

$$\nu(\lambda r)^{2\alpha} < \sqrt{A}\,\sin\varphi. \tag{4}$$

Equation (3) is the one ingredient of this paper that does not appear in the constructions themselves; Section 6 reports its verification against direct simulation of the nonlinear equations.

Two growth laws. The Boussinesq layers above grow at $\sqrt{A}$, a pendulum law: the amplitude pair obeys a second-order equation whose coefficient is the background gradient. The vortex layers of the hypodissipative Navier-Stokes construction instead grow at rate A_{n-1}, the background gradient itself, which is why its dissipation constraint reads $a = \alpha R$ in the exponent notation of Section 5 rather than carrying a square root. The swirl layers of the axisymmetric Euler construction obey the pendulum law, with the circulation gradient in place of the temperature gradient (Section 4.2). Both laws are treated below.

3 A bound independent of design constants

Theorem 3.1. *Consider a layer cascade with the following structure:*

(A1) each layer stops at amplitude $\lambda_q^{-1+\delta}$ with $\delta < 1$;

(A2) it deposits a background gradient $A_{q+1} = \lambda_q^{\delta}$, by (2);

(A3) frequencies and background gradients are related by $\lambda_q = A_q^p$ for a fixed p;

(A4) layers grow at rate A^γ with $\gamma = 1/2$ or $\gamma = 1$;

(A5) a layer of frequency λ is damped at $\nu\lambda^{2\alpha}$, as in (3).

Then $p\delta > 1$, and growth at every stage requires

$$\alpha \le \frac{\gamma}{2p} < \frac{\gamma\,\delta}{2} < \frac{\gamma}{2},$$

that is $\alpha < 1/4$ for $\gamma = 1/2$ and $\alpha < 1/2$ for $\gamma = 1$. In the $|\nabla|^\alpha$ convention these ceilings are $1/2$ and 1, against a classical viscosity of 2.

Proof. By (A2) and (A3), $A_{q+1} = \lambda_q^{\delta} = A_q^{p\delta}$. The gradients must grow without bound for the cascade to produce blowup, so $p\delta > 1$; with $\delta < 1$ from (A1), which is required because the field itself stays bounded while its gradient diverges, this gives $p > 1/\delta > 1$.

By (A5) and (A4), stage q grows only while $\nu\lambda_q^{2\alpha} < c\,A_q^\gamma$ for a constant c independent of q. Taking logarithms and using (A3),

$$\log\nu + 2\alpha p \log A_q < \log c + \gamma \log A_q,$$

and since $A_q \to \infty$ the inequality can hold for all large q only if $2\alpha p \le \gamma$, that is $\alpha \le \gamma/(2p)$. At equality the condition reduces to $\nu < c$, which a small viscosity satisfies, so this step is not strict; strictness of the final bounds comes from $p > 1/\delta$, which gives $\alpha \le \gamma/(2p) < \gamma\delta/2 < \gamma/2$. □

In the variables of Section 4, where frequencies satisfy $\lambda_q = \lambda_{q-1}^{Q}$, the exponents are related by $Q = p\delta$, and the bound $\alpha \le \gamma/(2p)$ reads $\alpha \le \gamma\delta/(2Q)$, which for $\gamma = 1/2$ is the constraint $\alpha \le m/(4Q)$ used there; the margin δ of (A1) is written m in that section, where δ denotes the per-level parameter of the correction hierarchy.

Corollary 3.2. *No cascade of either type reaches classical viscosity. The pendulum mechanism falls short by a factor of four and the vortex-stretching mechanism by a factor of two, uniformly in every constant of the design.*

The two ceilings differ by exactly the growth law: a layer that grows at the background gradient, rather than at its square root, gains a factor of two and nothing else in the argument changes. The published constructions sit well below their own class ceilings, the hypodissipative vortex-layer threshold by a factor 10.8 and the Boussinesq design of Section 4.1, at its published choices, by a factor 945.

Remark 3.3. Corollary 3.2 is consistent with a choice visible in the literature: the construction aimed at the viscous case does not use a layer cascade, but a self-similar core in which the scale-local Reynolds number stays of order one.

4 Bounds for the smooth-forcing designs

4.1 The Boussinesq design

The smooth-forcing Boussinesq construction [5] controls the force of each layer in $C^k_{x,t}$ for $k \le d = k_q$ through a finite hierarchy of corrections, and its estimates are powers of the layer frequency $\lambda = \lambda_q$. The inputs, all displayed in that paper, are: the amplitude $Y \le C\lambda^{-(1-m)}$, with margin $m = 1/8$; the frequency ratio $\lambda_q = \lambda_{q-1}^Q$ and the coefficient scale $\Pi \le \lambda^{5/Q}$; the per-level small parameter $\delta = \Pi^5 Y$; $J = 2d + 8$ correction levels, so that the uncancelled remainder begins at grade $J+1$; a cost of $\lambda\Pi^2$ per physical derivative of an oscillatory profile and of Π^2 per derivative of a phase-independent one; and coefficient derivatives through order $9d+41$, which are available uniformly only through order $Q-1$ because the sums over future stages converge only in that range. The force of one layer at order $k \le d$ then consists of the following contributions.

contribution	size
activation	$\Pi\lambda^{-d-6}(\lambda\Pi^2)^k$, set by the choice of seed
leading phase means	$\Pi^{2k+3}Y^2$
higher phase means	$\lambda Y\Pi^{2k-2}\delta^4$ and $Y\Pi^{2k-2}\delta^3$
remainders	$(\lambda\Pi^2)^k\,\lambda Y\,\delta^{2d+9}$

Each must be at most $\lambda^{-\tau}$ for some $\tau > 0$ for the force series to converge. The construction takes $\tau = 3/2$, which absorbs its constants into a logarithm, together with the sufficient rules $Q \ge 201$ and $120d \le Q$.

Theorem 4.1. *Write $\pi = 5/Q$. The estimates behind the table hold uniformly only if $Q \ge 9d+42$, and its bounds are at most $\lambda^{-\tau}$ for all large λ only if*

$$m \le \frac{d+9-2d\pi-5\pi(2d+9)-\tau}{2d+10}, \qquad m \le 1-\frac{(2d+3)\pi+\tau}{2}, \qquad m \le \frac{4-\tau-(2d+18)\pi}{5},$$

and carrying dissipation $-\nu(-\Delta)^\alpha$ requires in addition $\alpha \le m/(4Q)$. Consequently, in our convention:

(i) at the published choices $\tau = 3/2$, $Q \ge \max(201, 120d)$, the exponent satisfies $\alpha \le 2.65\times10^{-4}$, attained at $d = 0$, $Q = 201$, where the leading phase means bind;

(ii) with every free choice released, $\tau \to 0$ and only $Q \ge 9d + 42$, it satisfies $\alpha \le 2.22 \times 10^{-3}$, attained at $d = 0$, $Q = 45$;

(iii) even with the coefficient scale reduced to the previous wavelength, $\Pi \le \lambda^{1/Q}$, and the per-level parameter to $\Pi^3 Y$, the least the correction estimates tolerate, it satisfies $\alpha \le 4.48 \times 10^{-3}$.

In each case the exponent is largest when the fewest derivatives are controlled, and it is at least ten times below the threshold 0.04633 *already reached with a force of finite regularity.*

Proof. Taking the exponent of λ in each row and requiring it to be at most $-\tau$ gives one linear inequality in m per row; every exponent increases with k, so $k = d$ is the binding order. The remainder row gives the first display, the leading means the second, the vorticity higher means the third; the scalar higher means and the requirement that each level gain, $m < 1 - 25/Q$, are weaker in every case considered. The activation row is met by the choice of seed and constrains nothing else. The condition $Q \geq 9d + 42$ is the requirement that order $9d + 41$ lie within the range $Q - 1$. The dissipation condition is (4) with $A_q = \lambda_{q-1}^m$ and an insertion angle of order one, the most favourable case, which is Theorem 3.1 with $p = Q/m$ and $\gamma = 1/2$. The three consequences follow by maximizing $m/(4Q)$ over $d \geq 0$ and integer Q above each floor. As a check on the transcription, the published margin $m = 1/8$ satisfies all three displayed inequalities at the published choices for every d, and at $\tau = 3/2$ the second inequality is the construction's own condition $\Pi^{2d+3} \leq \lambda^{1/4}$. □

	published choices			free choices released		
d	Q	$m \leq$	$\alpha \leq$	Q	$m \leq$	$\alpha \leq$
0	201	0.213	2.65×10^{-4}	45	0.400	2.22×10^{-3}
1	201	0.188	2.34×10^{-4}	57	0.417	1.83×10^{-3}
2	240	0.177	1.84×10^{-4}	63	0.395	1.57×10^{-3}
5	600	0.196	8.16×10^{-5}	87	0.398	1.14×10^{-3}
10	1200	0.202	4.21×10^{-5}	132	0.425	8.05×10^{-4}

In the $|\nabla|^\alpha$ convention the three ceilings are 5.29×10^{-4}, 4.44×10^{-3} and 8.96×10^{-3}, against a proved threshold of 0.0927: factors of 175, 21 and 10. What ties the regularity of the force to the frequency ratio is the derivative range $Q \geq 9d + 42$, and what caps the margin is the correction hierarchy itself: two levels per derivative, each gaining $\lambda^{-(1-m)}$, drive the remainder bound on m down to $1/2$ as d grows, and at the published target the leading phase means, which are quadratic in the amplitude and gain nothing from the hierarchy, hold it near $1/4$.

4.2 The Euler design

In the axisymmetric Euler construction [6], written in the volume coordinates $y = (z, r^2/2)$, the circulation $\Gamma = ru^\varphi$ takes the place of the temperature. The principal amplitudes of a layer of frequency N obey

$$\partial_t \begin{pmatrix} T \\ \widehat{\Omega} \end{pmatrix} = \begin{pmatrix} 0 & -d/(\sigma\kappa) \\ \sigma c\, \zeta_1 & 0 \end{pmatrix} \begin{pmatrix} T \\ \widehat{\Omega} \end{pmatrix}, \qquad d = (J\zeta) \cdot \nabla_y \Gamma_<, \qquad c = 2W\Gamma_<,$$

with c bounded above and below on the support of the layer. The growth rate is the square root of the product of the off-diagonal entries and scales as $|\nabla\Gamma|^{1/2}$: the Euler layers belong to the pendulum class of Theorem 3.1, with $\gamma = 1/2$ in the variable $\nabla\Gamma$ whose norm diverges. On the constructed sequence the growth scale is $\sigma_i \asymp N_i^{\beta/2}$ against a deposited circulation gradient of order N_i^β, in agreement with this law.

The force estimates of this construction are stated with explicit exponents. Write β for the amplitude margin (amplitude $N^{-(1-\beta)}$), $h = 1 - \beta$, $N_i = N_{i-1}^Q$, $\Pi \leq N^v$ with $v = p/Q$ for the coefficient scale, $\delta = \Pi^{\mathsf{d}} N^{-h}$ for the per-level parameter, $h_* = h - \mathsf{d}v$, $J = 2k + 8$ correction levels

and $u \geq kv$. The activation, remainder and mean contributions to the force of one layer at order k then carry the exponents

$$1 - e + 2u + v, \qquad -(2h_* - 1)k + \beta - 9h_* + 2u + v, \qquad -2h + 3u + 6v$$

of N, where e is the exponent of the seed. The estimates further require thirteen coefficient quotients of the form $\Pi^{A-B\mathsf{d}}N^{-n\beta}$ to be at most one, the localization exponent a of $\ell = N_{i-1}^{-a}$ to satisfy $a > 5\beta/2$ with $p \geq a$, the centre condition $2h_* > \beta/2$, and coefficient derivatives through the orders the correction recursion consumes, which are available uniformly only when $9k+51 \leq Q$. The construction chooses $\beta = 1/8$, $\mathsf{d} = 16$, $p = 8$, $a = 4$ and $Q_i = q_0 + i$ with $q_0 \geq 2^{20}$; with these choices the three exponents are -7.00, -10.75 and -1.75, within its targets -6, -6 and $-3/2$, and every side condition holds.

Theorem 4.2. *Under the Euler design, carrying dissipation $-\nu(-\Delta)^\alpha$ at an insertion angle of order one requires $\alpha \leq \beta/(4Q)$. Consequently, in our convention:*

(i) at the published choices $\alpha \leq 2.98 \times 10^{-8}$ at the first stage, and no positive exponent is admissible at every stage, since Q_i is unbounded;

(ii) with every free choice released, that is, force exponents only required to be negative, d at the least value the coefficient quotients permit and $Q \geq 9k + 51$, the exponent satisfies $\alpha \leq 3.64 \times 10^{-3}$, attained at $k = 0$, $Q = 51$, $\beta = 0.742$, where the centre condition binds;

(iii) with the localization at the previous wavelength, $a = 1$, it satisfies $\alpha \leq 3.77 \times 10^{-3}$.

Proof. Growth at frequency N_i on the circulation gradient deposited by the previous layer requires $\nu N_i^{2\alpha} < \sigma_{i-1} \sin s_i$ with $\sigma_{i-1} \asymp N_{i-1}^{\beta/2}$ and $N_i = N_{i-1}^Q$, which is $\alpha < \beta/(4Q)$; this is Theorem 3.1 with $p = Q/\beta$ and $\gamma = 1/2$. The first consequence is this bound at $Q = 2^{20} + 1$ and $\beta = 1/8$. For the others, each listed condition is an inequality in β at fixed k and Q; d is taken as the least value for which every coefficient quotient is at most one, which is $\mathsf{d} = 2$ in the maximizing range because the quotients carrying no power of N are $\Pi^{2-\mathsf{d}}$; and $\beta/(4Q)$ is maximized over $k \geq 0$ and integer $Q \geq 9k + 51$. □

In the $|\nabla|^\alpha$ convention the ceilings of (ii) and (iii) are 7.27×10^{-3} and 7.53×10^{-3}, below the proved threshold 0.0927 by factors of 12.7 and 12.3; with one controlled derivative of the force, (ii) gives 3.12×10^{-3}. The two smooth-forcing designs therefore behave alike: both are of the pendulum class, both tie the frequency ratio to the regularity of the force through the range of their coefficient bounds, $Q \geq 9d + 42$ and $Q \geq 9k + 51$, and both certify at most a few thousandths, more than ten times below the rough-force threshold. For a Navier-Stokes version of the Euler design the damping of a localized swirl layer is taken at its leading symbol $\nu(N|\zeta|)^{2\alpha}$; the verification of Section 6 concerns the Boussinesq system.

Remark 4.3. A dissipative version of either design must therefore change one of three things: the range of uniform coefficient bounds, which is what forces Q to grow with d; the treatment of the nonoscillatory phase means, which no number of correction levels reduces; or the growth law, since the factor four in $\alpha \leq m/(4Q)$ comes from the rate $\sqrt{A}$ and becomes a factor two for vortex stretching. The bounds concern what the estimates of each design can certify, not the existence of any solution: a different correction scheme is not covered, and Theorem 3.1 is the only statement here independent of the designs.

5 The exponent budget behind the published threshold

The hypodissipative Navier-Stokes construction [3] writes every quantity as a power of one base: the frequency of layer n is $M_n = N^{R^n}$, the amplitude $A_n = N^{aR^n}$, the localization scale $L_n = N^{bR^n}$, and $s > 0$ is the Hölder margin of the force. Its force estimates give four constraints:

	origin	constraint
D	dissipation	$a = \alpha R$
S	self-interaction	$2a - a/R + b + s - 1 \le 0$
L	localization	$b \ge a + 1/R + s$
O	outer velocity on the inner layer	$a + 2/R + s + 1 - 3b \le 0$

Proposition 5.1. *With **D** and **S** saturated, **O** reduces to $4s < 2 + 3\alpha - 7\alpha R - 2/R$, whose maximum over the frequency ratio is at $\alpha R^2 = 2/7$. At that optimum $s(\alpha) = (2+3\alpha-2\sqrt{14\alpha})/4$, and $s > 0$ is equivalent to $9\alpha^2 - 44\alpha + 4 > 0$, whose smaller root is $(22 - 8\sqrt{7})/9$.*

The optimizing ratio $R = \sqrt{2/(7\alpha)}$ is the one the construction uses, and no coefficient in the chain is fitted. At the threshold the localization constraint is inactive, with margin $\frac{1}{2}(\sqrt{2\alpha/7}-\alpha) = 0.0350$; a budget that saturates **L** instead and omits **O** yields $\alpha R^2 = 1/3$ and the weaker bound $5-2\sqrt{6} = 0.10102$, the value given by an order-of-magnitude estimate in the source. The difference between the two is precisely which constraint is active.

Proposition 5.2. *A positive force margin requires $7\alpha R^2 - (2 + 3\alpha)R + 2 < 0$, so the admissible ratios form the interval with endpoints $R_\pm = \left[(2+3\alpha) \pm \sqrt{9\alpha^2 - 44\alpha + 4}\right]/(14\alpha)$. The discriminant is the polynomial of Proposition 5.1, so the interval closes to a single point exactly at the threshold, and that point is $\sqrt{2/(7\alpha)}$. Moreover $R_- > 1$ for every $\alpha > 0$, and at $R = 1$ the margin equals $-\alpha$.*

α ($\lvert\nabla\rvert^\alpha$)	admissible R	optimizing R
0.001	(1.002, 285.1)	16.90
0.01	(1.021, 27.98)	5.345
0.05	(1.143, 5.000)	2.390
0.09267	{1.7559}	1.7559
0.10	empty	none

Corollary 5.3. *Cascades with geometrically growing frequencies, the case $R = 1$, are inadmissible at every positive dissipation: localization requires $b \ge a + 1 + s$ while self-interaction permits only $b \le 1 - a - s$, forcing $\alpha + s \le 0$. Super-geometric frequency growth is therefore necessary, and more of it is necessary as the dissipation rises, until at the threshold a single ratio remains.*

Propositions 5.1 and 5.2 are verified in exact arithmetic by computer algebra in the accompanying repository.

6 Verification of the dissipative extension

Theorems 3.1, 4.1 and 4.2 rest on (3), which is derived rather than taken from the constructions. It was tested against direct pseudospectral simulation of the nonlinear two-dimensional Boussinesq system in double precision with two-thirds dealiasing.

The amplitude system. On the stratification $\theta = -(A/\lambda_0)\sin(\lambda_0 x_2)$ the measured peak growth rate matches $\sqrt{A}\sin\varphi$ to 4×10^{-5} relative at two insertion angles; the predicted independence of the rate from λ holds to 2.7×10^{-4} across an eightfold range; and the damping term of (3) tracks the measured rate to 2.6×10^{-4} absolute across $\alpha\in\{1/4,1/2,1\}$ and $\nu\in\{10^{-5},10^{-4}\}$.

The cascade premise. On a two-scale background that is an exact steady state, with drift 1.45×10^{-15}, the local growth rate of a fine wave follows the total gradient at correlation 0.99945 and the base stratification alone at 0.212: a layer responds to the accumulated low-frequency gradient rather than to the base.

The growth, steering and holding cycle. The control that returns a grown layer to rest rotates the background gradient and the phase direction together, so that the laboratory component ζ_1 changes sign while $J\zeta\cdot G$ and $|\zeta|$ are preserved. In the co-rotating frame it becomes a rotating gravity direction, with the stratification held by a low-frequency force. On a background whose gradient is affine to fourth order the cycle reproduces the amplitude system to 3.06×10^{-6} in the steering gain and returns the vorticity amplitude to 3.67×10^{-4} of its peak, where it holds to 0.36% over three growth times; on a plain sinusoidal background the same quantities are 1.80×10^{-2} and 1.57×10^{-2}, the difference being the curvature of the background over the layer's support. Under dissipation the cycle factorizes as e^{-dt} times the inviscid cycle with $d=\nu\lambda^{2\alpha}$: a holding interval decays at d to within 0.97%, and the rescaled viscous run reproduces the inviscid one to 0.24%.

Localization. Since the constructions localize each layer, (3) is applied to a band rather than to a single frequency. For an envelope of radius ℓ on a wave of frequency λ the relative error of the damping law is

$$\frac{\big\|(-\Delta)^{\alpha}(g\,e^{i\lambda\zeta\cdot x})-(\lambda|\zeta|)^{2\alpha}g\,e^{i\lambda\zeta\cdot x}\big\|}{\big\|(\lambda|\zeta|)^{2\alpha}g\big\|}\;\approx\;\frac{3.6\,\alpha}{\ell\lambda},$$

first order in the bandwidth ratio, with measured slope -0.9863, and depending on that ratio alone: configurations sharing $\ell\lambda$ but splitting it differently between frequency and envelope agree to four digits. The coefficient is linear in α, as the symbol expansion predicts. At the separations the constructions use, $\ell_q^{-1}=\lambda_{q-1}^{3}$ against $\lambda_q=\lambda_{q-1}^{Q_q}$ with $Q_q\geq201$, the error is $\lambda_{q-1}^{3-Q_q}$, so localization does not affect the bounds above.

A property of the mechanism in an unforced periodic setting. The stratification these layers grow on is Rayleigh-Taylor unstable at rate $\sqrt{A}$, while a layer inserted at angle s grows at $\sqrt{A}\sin s$. Perturbations at more favourable angles therefore gain $1/\sin s$ times as many e-foldings as the layer. The constructions are unaffected, since each layer is localized and the force controls the exterior, but any realization without that control must keep its schedule short enough for the layer to stay ahead.

7 The status of the announced results

The two September 2026 announcements [8, 9] are accompanied by machine-checked certificates. For the Navier-Stokes and Euler development, the headline statements match the forced alternatives of the official problem statement [1] clause by clause; the development builds with no incomplete proofs and its four headline theorems depend only on the three standard axioms; and

every constant in the closure of both root modules replays through the Lean kernel from an empty environment at the pinned toolchain version. The reference statement is adapted from an independently maintained collection of formalized conjectures, so the definitions that determine whether the formalized statement is the intended one predate the claim. None of this speaks to the mathematics outside the formalization, and kernel replay detects environment manipulation rather than serving as an external verifier.

These checks concern the formal statements and the integrity of the formal proof. They do not establish that the formal proof follows the written argument. Bastounis, Circelli and Hansen [2] report that the Lean proof of the Navier-Stokes announcement does not correspond to its natural-language proof, exhibiting places where the text states a stronger bound than the cited Lean declaration proves; they do not claim the written argument is false. Their finding is consistent with ours: a formal proof of a faithful statement establishes that statement, while the correctness of the written argument, and of any result that cites it rather than the formal development, remains a separate question.

8 Scope

The bound of Section 3 is a necessary condition for growth in the layer mechanism, derived from the amplitude system, the amplification identity and the dissipative term (3). The bounds of Section 4 concern what the estimates of two designs can certify: they track each design's corrections through the exponents of its published estimates, but not constants or logarithmic factors, and they assume the most favourable insertion angle. Neither bound affects the published inviscid results, which carry no dissipation. The mechanism is due to Córdoba and Martínez-Zoroa [4, 3], extended to smooth forcing by Alpogé and Buckmaster [5, 6, 7]; nothing in this paper establishes blowup for any equation or improves any published threshold.

Evaluating the relation $\alpha_c = 1/(4p)$ at the published threshold gives $p = 11/4 + \sqrt{7}$. This value carries no information about the threshold: the relation is constraint **D** saturated, which the construction saturates at every α, so the identity holds along the whole family, and its algebraic form is automatic because $\alpha_0 \in \mathbb{Q}(\sqrt{7})$ and the norm of $22 - 8\sqrt{7}$ is a perfect square.

Data and code

The numerical data, the simulations, the exact-arithmetic verifications of Propositions 5.1 and 5.2, and the computations behind Theorems 3.1, 4.1 and 4.2 are public at the project repository.

References

[1] C. L. Fefferman, *Existence and smoothness of the Navier-Stokes equation*, Clay Mathematics Institute Millennium Prize problem description.

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